

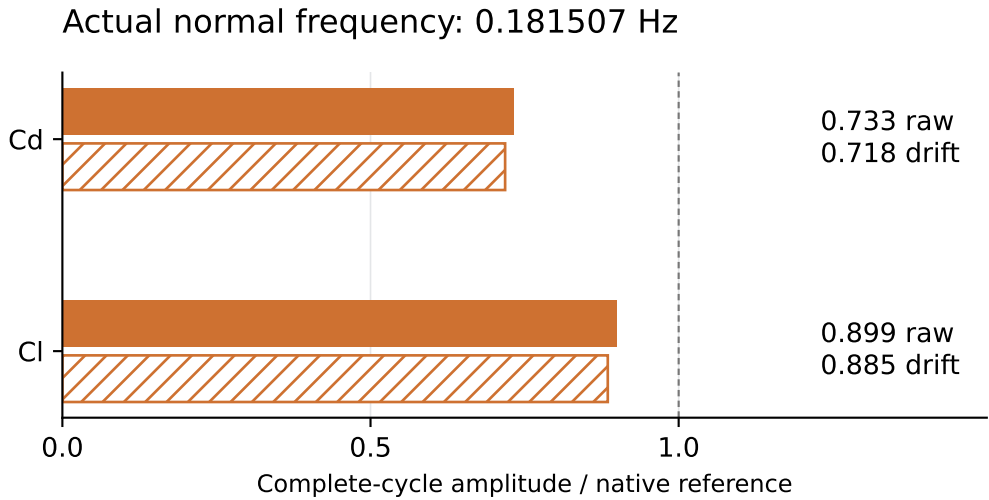
Normal-velocity spatial modes and frequency

2 × 2 forced FVM control; complete force cycles wholly within 102–112 s

Actual DC/trend, tangential velocity and Gt are fixed; each mode keeps its initial force-relative phase.

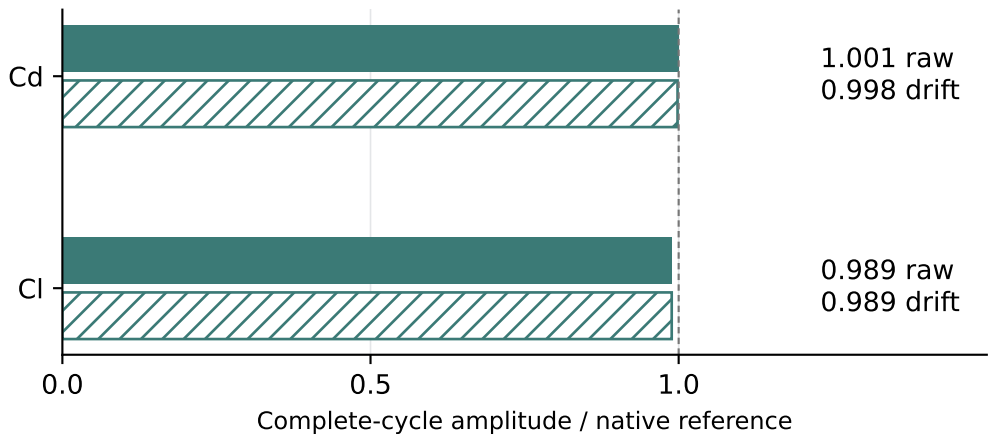
Mode = measured spatial fundamental/second-harmonic coefficients. Filled: raw; hatched: drift corrected.

Actual Un modes



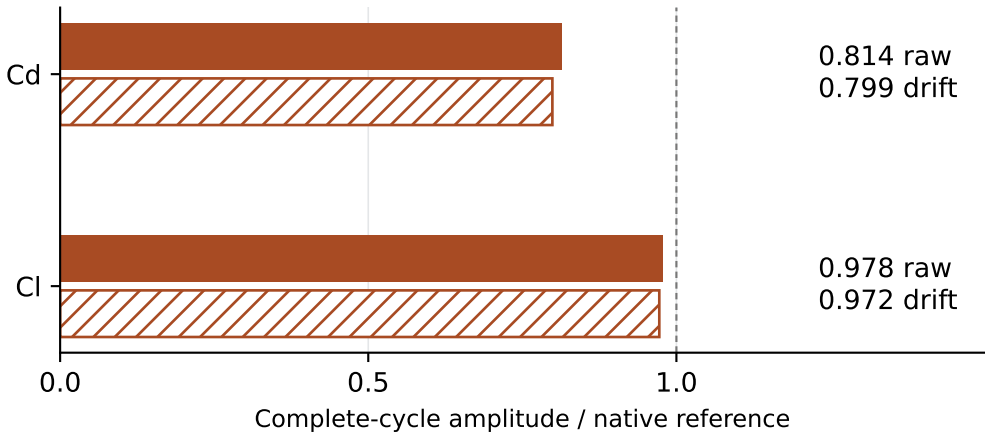
2 Cd cycles · 1 Cl cycle; last/first Cd amplitude 0.912

Reference Un modes

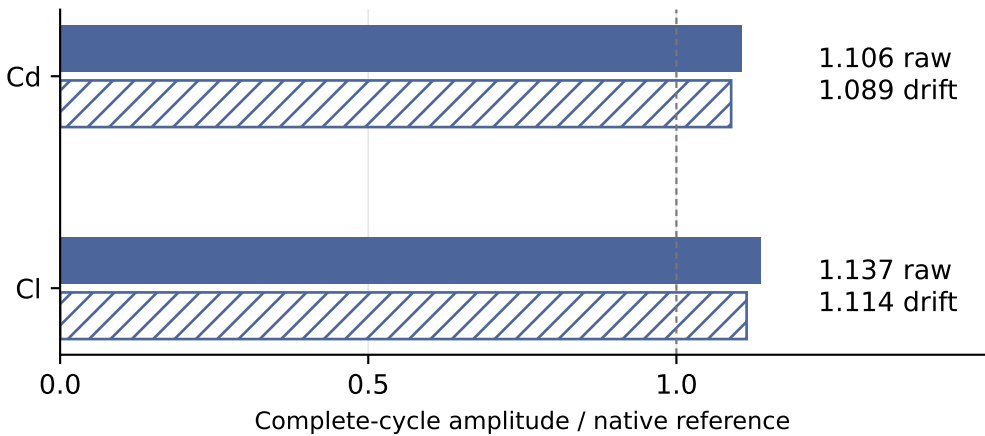


2 Cd cycles · 1 Cl cycle; last/first Cd amplitude 1.004

Reference normal frequency: 0.187330 Hz



2 Cd cycles · 1 Cl cycle; last/first Cd amplitude 1.007



3 Cd cycles · 1 Cl cycle; last/first Cd amplitude 1.102

Frequency substitutions change only Un oscillator rate; its subsequent phase relative to unchanged Ut/Gt evolves.

Two drag cycles versus three in the last cell; growth and one lift cycle limit saturation claims.

Common native mapped initialization and numerical settings. These responses do not demonstrate autonomous recovery.